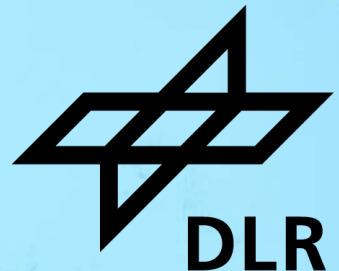
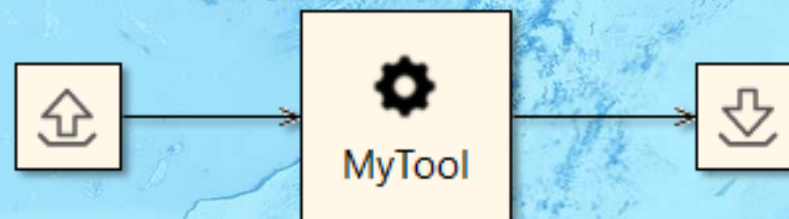


RCE TOOL INTEGRATION

Dummy Tool Wrapper

DLR Institute for System Architectures in Aeronautics



 Integrate a Tool as a Workflow Component

Choose Tool Configuration

Create a new tool configuration or choose an inactive one to activate

☒ Create a new Common tool configuration
☐ Create a new CPACS tool configuration
☐ Create a new tool configuration from a template

CPACS Tool (Type: CPACS)
CPACS Tool with incoming and return directory (Type: CPACS)
CPACS Tool with return directory (Type: CPACS)

☐ Choose an inactive tool configuration to edit:

Selected tool configuration:



(1) In this example we use the "Common tool configuration".

(2) Click "Next >"

Integrate a Tool as a Workflow Component

Inputs and Outputs

Configure the inputs and outputs of the tool

Inputs Outputs Verification

Input	Data type	Handling	Constraint
XML_in	File	Constant (not consumed), S...	Required

Add... Edit... Remove

Edit... Input

Name*: XML_in

Data type*: File

Handling*: ☒ Constant (not consumed)
☒ Single (consumed)
☒ Queue (consumed)

Default handling*: Queue (consumed)

Constraint*: ☒ Required
☐ Required if connected
☐ Not required

Default constraint*: Required

OK Cancel

< Back Next > Save As ... Save and activate Cancel

(4) Outputs

(1) Click "Add"

(2) Define RCE input variable of type "File"

(3) Choose the file handling settings and click "OK"

R Integrate a Tool as a Workflow Component

Inputs and Outputs

Configure the inputs and outputs of the tool

Inputs Outputs Verification

Output	Data type
XML_out	File

Add...
Edit...
Remove

R Edit... Output

Name*: XML_out

Data type*: File

OK Cancel

(1) Click "Add"

(2) Define RCE output variable of type "File"

(3) Next >

? < Back Next > Save As ... Save and activate Cancel

Tool Properties

Define properties of the tool

Property groups:

Default

Add...

Rename...

Remove

Property keys for group "Default":

Key	Display name	Define at wo...	Default va...	Comment

Add...

Edit...

Remove

Nothing to do ...

☐ Write key-value properties file at runtime (in "Config" folder)

working/Config/

(1) Next >



< Back

Next >

Save As ...

Save and update

Cancel



Integrate a Tool as a Workflow Component

Launch Settings

⚠ Currently, only one launch setting is possible

Host	Tool directory	Version	Working directory
RCE	D:\Entwicklung\CPACS\cpa...		

Add...

Add Launch Settings

Tool directory*: D:\Entwicklung\CPACS\cpacs_seminar\ToolInte ...

Version*: 1.0

Working directory (absolute): ...

☒ Create arbitrary directory in RCE temp directory

☐ Limit parallel executions 10

OK Cancel

☐ Use a new working directory on each run

Tool Copying Behaviour

☒ Do not copy tool

☐ Copy tool to working directory once

☐ Copy tool to working directory on each run

Clean up choices

☐ Never delete working directory(ies)

☒ Delete working directory(ies) when workflow is finished

☒ Keep in case of failed workflow run

☐ Delete working directory(ies) after each run of the tool

☐ Keep in case of failed tool run

*Defines the user's choices when configuring the component

(1) Add launch setting

(2) Point to tool directory

(3) Specify tool version

(4) Select checkboxes like this

(5) Select your preferred copying behavior

(6) Good for debugging...

(7) Next >

< Back Next > Save As ... Save and activate Cancel


```
C:\ProgramData\mambaforge_24.3.0.0\Scripts\activate.bat cpacs
```

```
python "${dir:tool}"/main.py --cpacs_input="${in:XML_in}" --cpacs_output="cpacs_out.xml"
```



(1) Specify execution commands for Windows and/or Linux



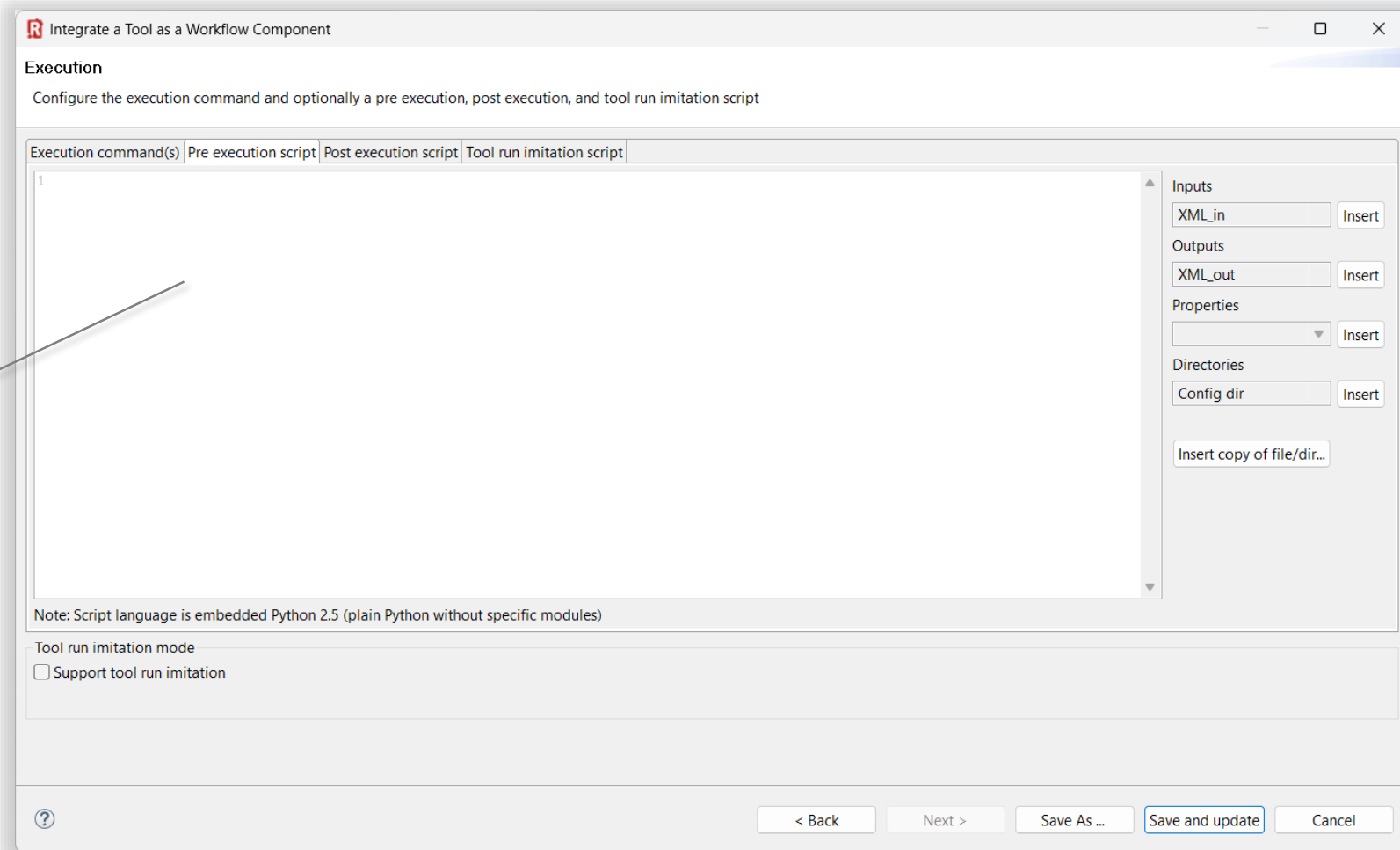
This example is written in Python, so we need to activate the correct interpreter first.

(2) Select "Working directory"

The screenshot shows the 'Integrate a Tool as a Workflow Component' dialog box. The 'Execution' tab is active, showing fields for 'Execution command(s)', 'Pre execution script', 'Post execution script', and 'Tool run imitation script'. The 'Execution command(s)' field contains a Windows batch command to activate the Python environment and run a script. The 'Pre execution script' field is empty. The 'Post execution script' field is empty. The 'Tool run imitation script' field is empty. The 'Command(s) for Windows' checkbox is checked, and the 'Command(s) for Linux' checkbox is unchecked. The 'Note: Command(s) executed as batch file' is displayed. The 'Note: Command language is Bash' is displayed. The 'Execution Options' section has the 'Exit code other than 0 is not an error' checkbox unchecked. The 'Execute (command(s), pre execution/post execution/tool run imitation script) from' section has the 'Working directory' radio button selected. The 'Tool run imitation mode' section has the 'Support tool run imitation' checkbox unchecked. The 'Inputs' section has an 'XML_in' input field with an 'Insert' button. The 'Properties' section has an empty input field with an 'Insert' button. The 'Directories' section has a 'Config dir' input field with an 'Insert' button. The bottom of the dialog has buttons for '< Back', 'Next >', 'Save As ...', 'Save and update', and 'Cancel'.

(1) Insert pre-execution script in Python

💡 Pre-execution: What happens before the actual tool is activated. This is useful if you need to prepare a specific folder structure or copy resources before execution.



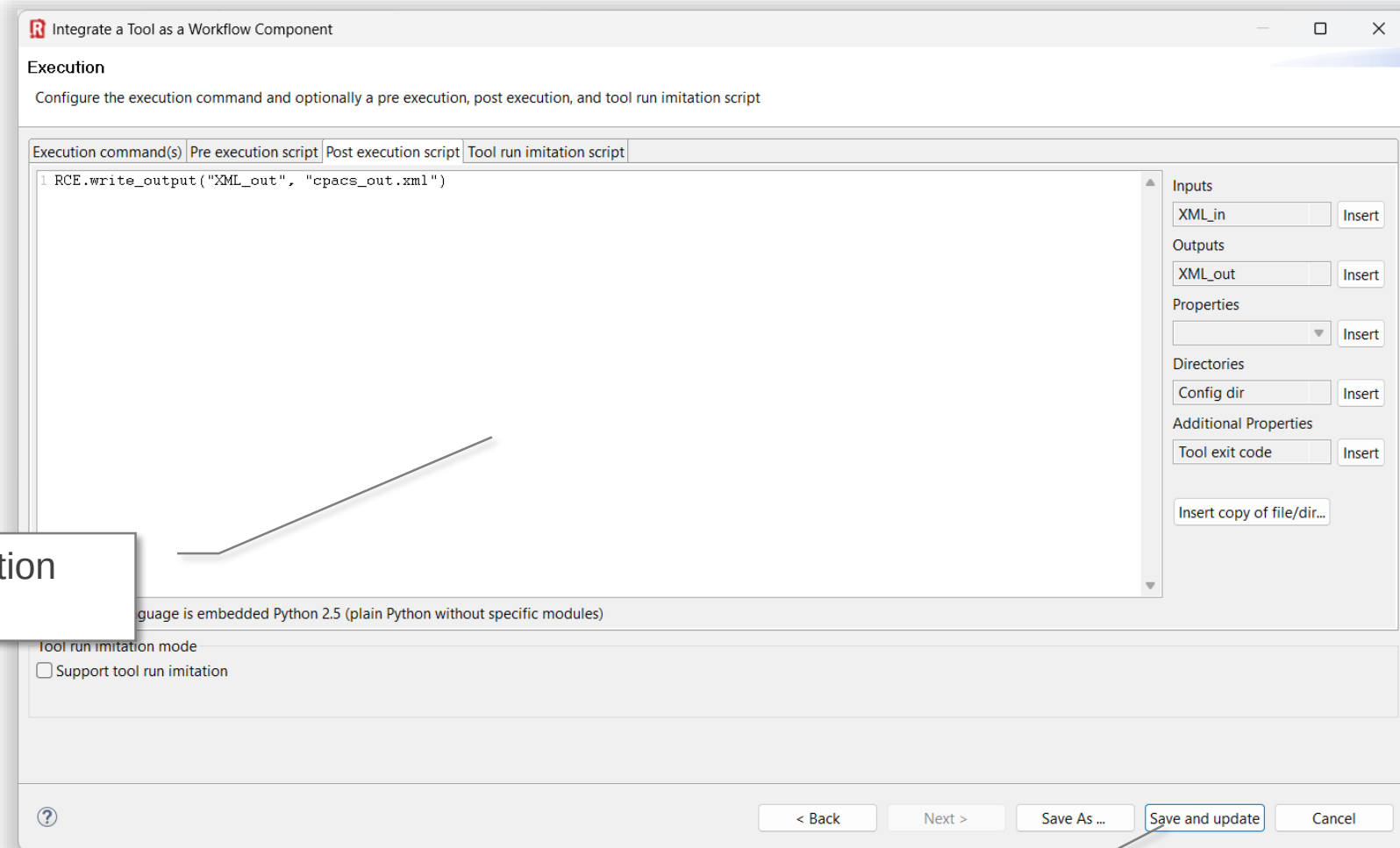
The screenshot shows a software interface window titled "Integrate a Tool as a Workflow Component". The "Execution" tab is selected, with the instruction "Configure the execution command and optionally a pre execution, post execution, and tool run imitation script". The interface features four tabs: "Execution command(s)", "Pre execution script", "Post execution script", and "Tool run imitation script". The "Pre execution script" tab is active, showing a large text area with a single line of text "1". To the right of the text area is a sidebar with sections for "Inputs" (XML_in), "Outputs" (XML_out), "Properties", "Directories" (Config dir), and an "Insert copy of file/dir..." button. Each section has an "Insert" button. At the bottom of the window, there is a "Note: Script language is embedded Python 2.5 (plain Python without specific modules)", a "Tool run imitation mode" section with a checked "Support tool run imitation" checkbox, and a row of navigation buttons: "< Back", "Next >", "Save As ...", "Save and update", and "Cancel".

```
RCE.write_output("XML_out", "cpacs_out.xml")
```



Post-execution: What happens after the actual tool has finished. Here we're telling RCE where to find the output file.

(1) Insert post-execution script in Python



Integrate a Tool as a Workflow Component

Execution

Configure the execution command and optionally a pre execution, post execution, and tool run imitation script

Execution command(s) Pre execution script Post execution script Tool run imitation script

RCE.write_output("XML_out", "cpacs_out.xml")

Inputs

XML_in Insert

Outputs

XML_out Insert

Properties

Insert

Directories

Config dir Insert

Additional Properties

Tool exit code Insert

Insert copy of file/dir...

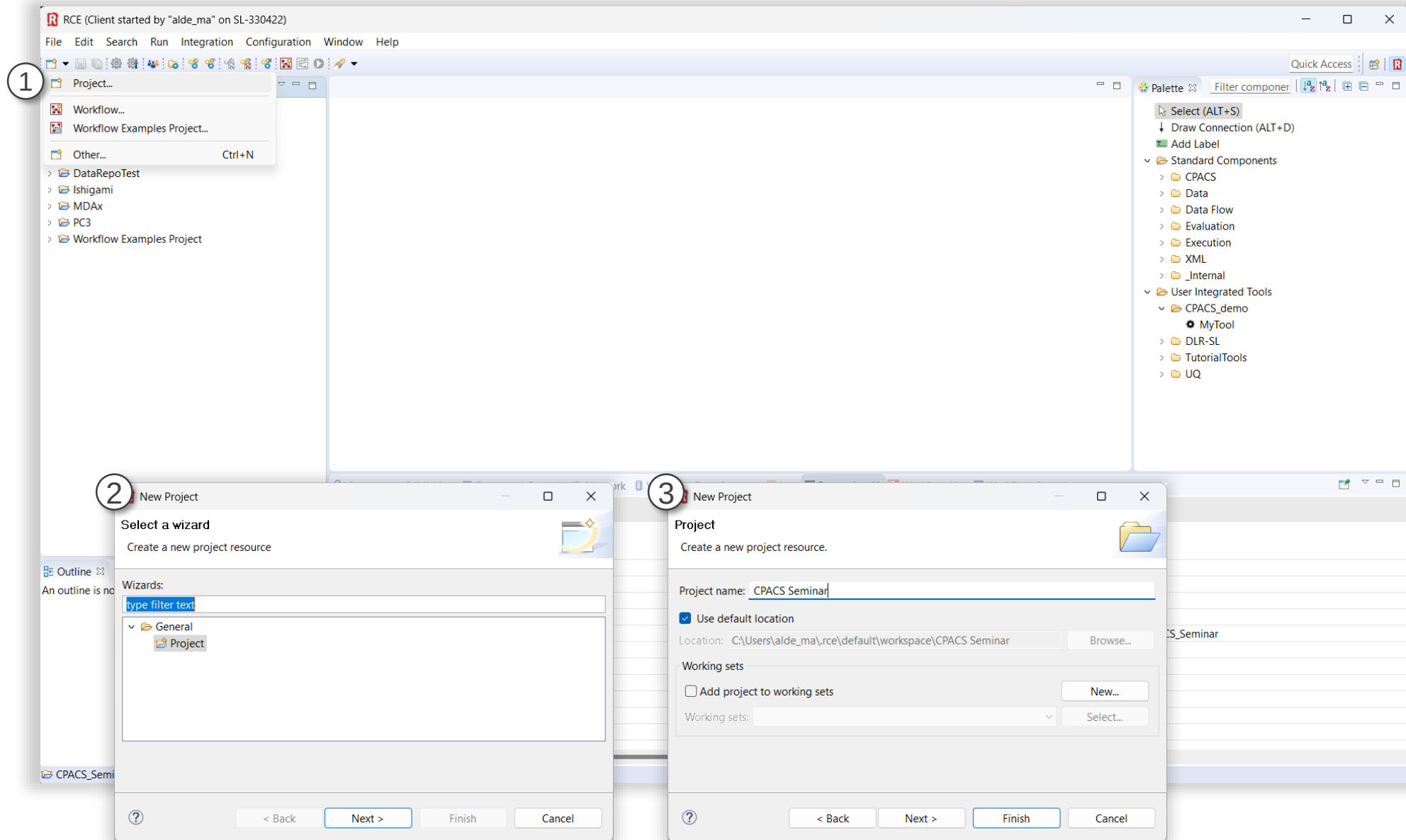
Language is embedded Python 2.5 (plain Python without specific modules)

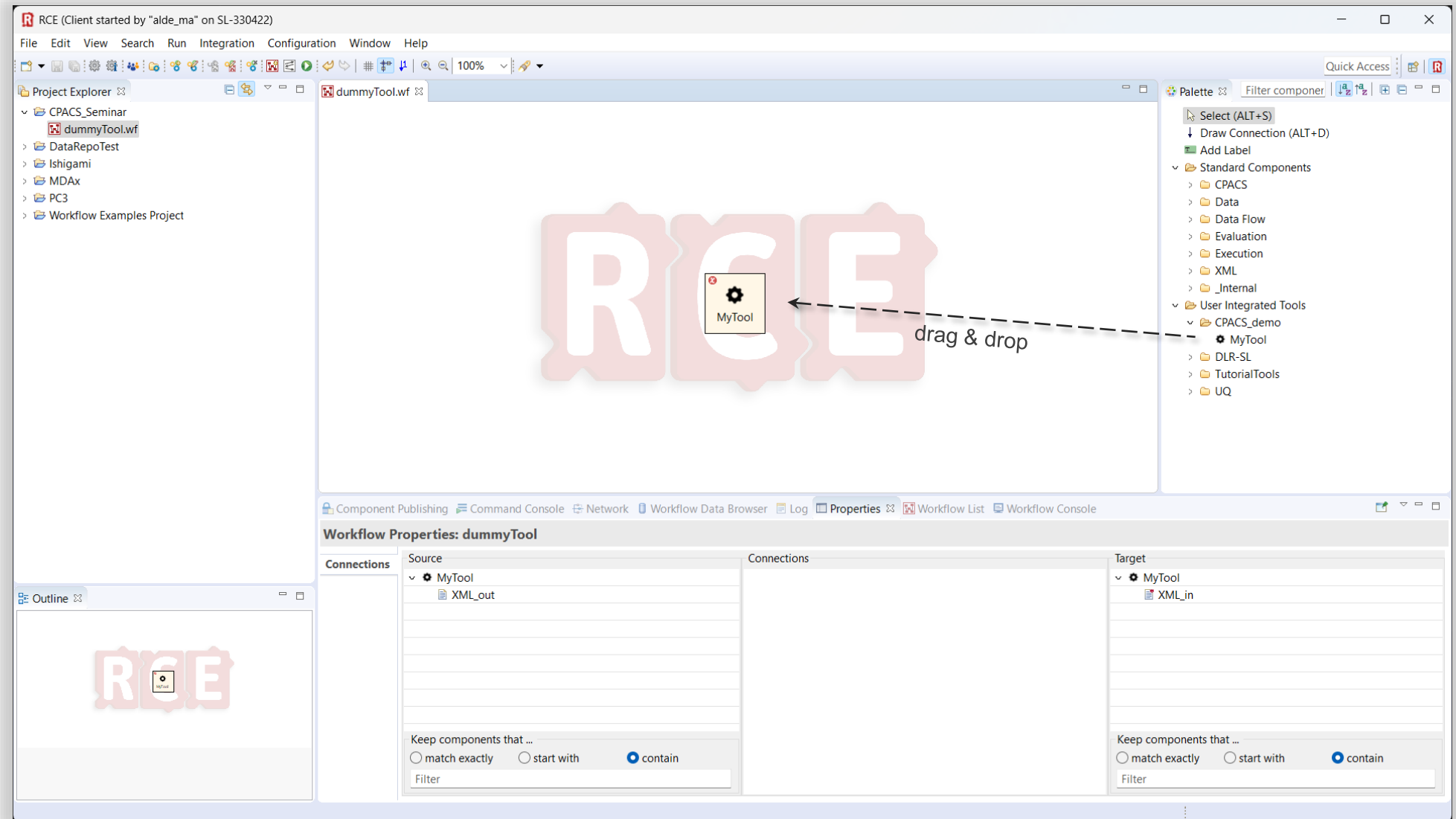
Tool run imitation mode

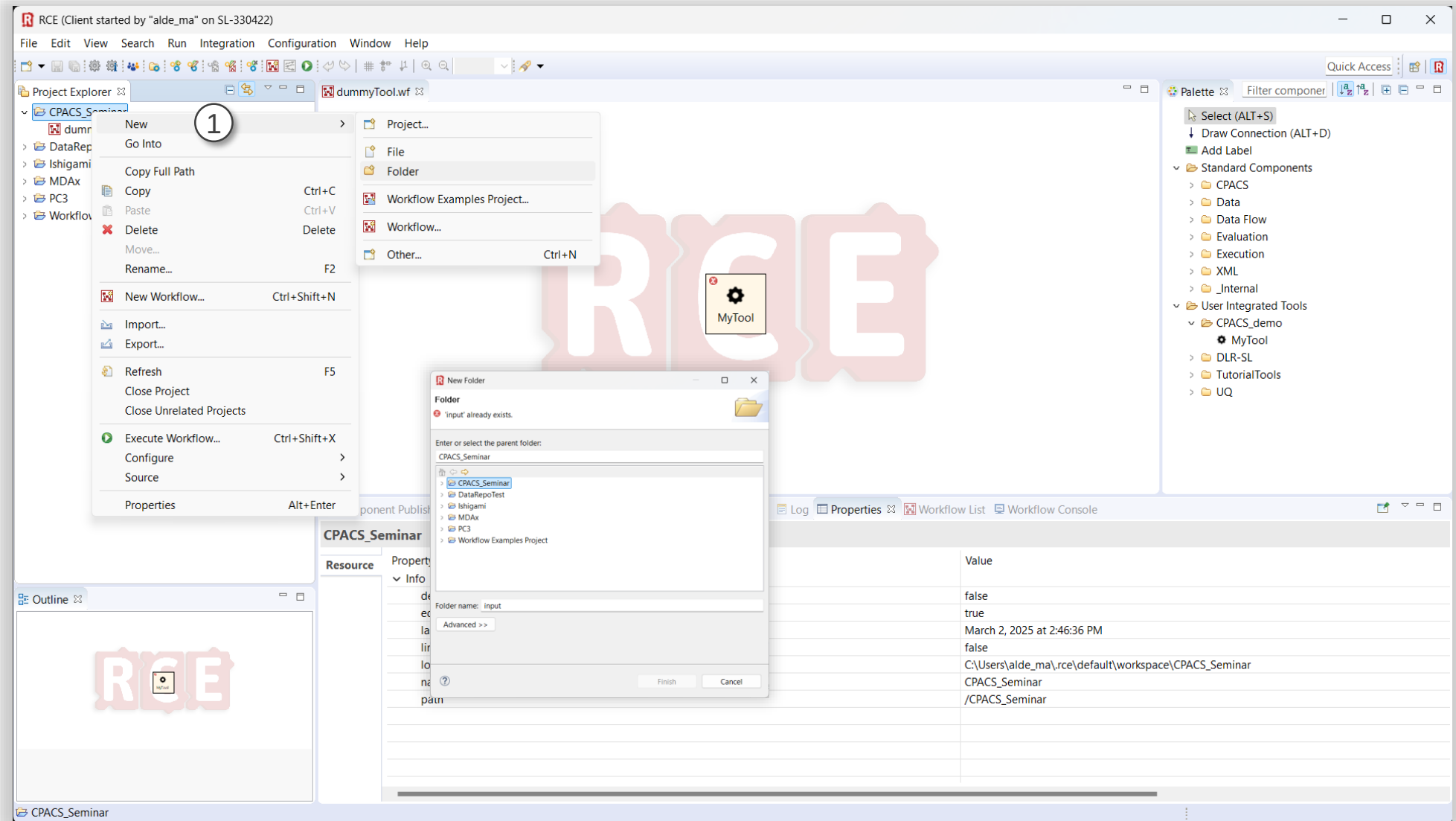
☐ Support tool run imitation

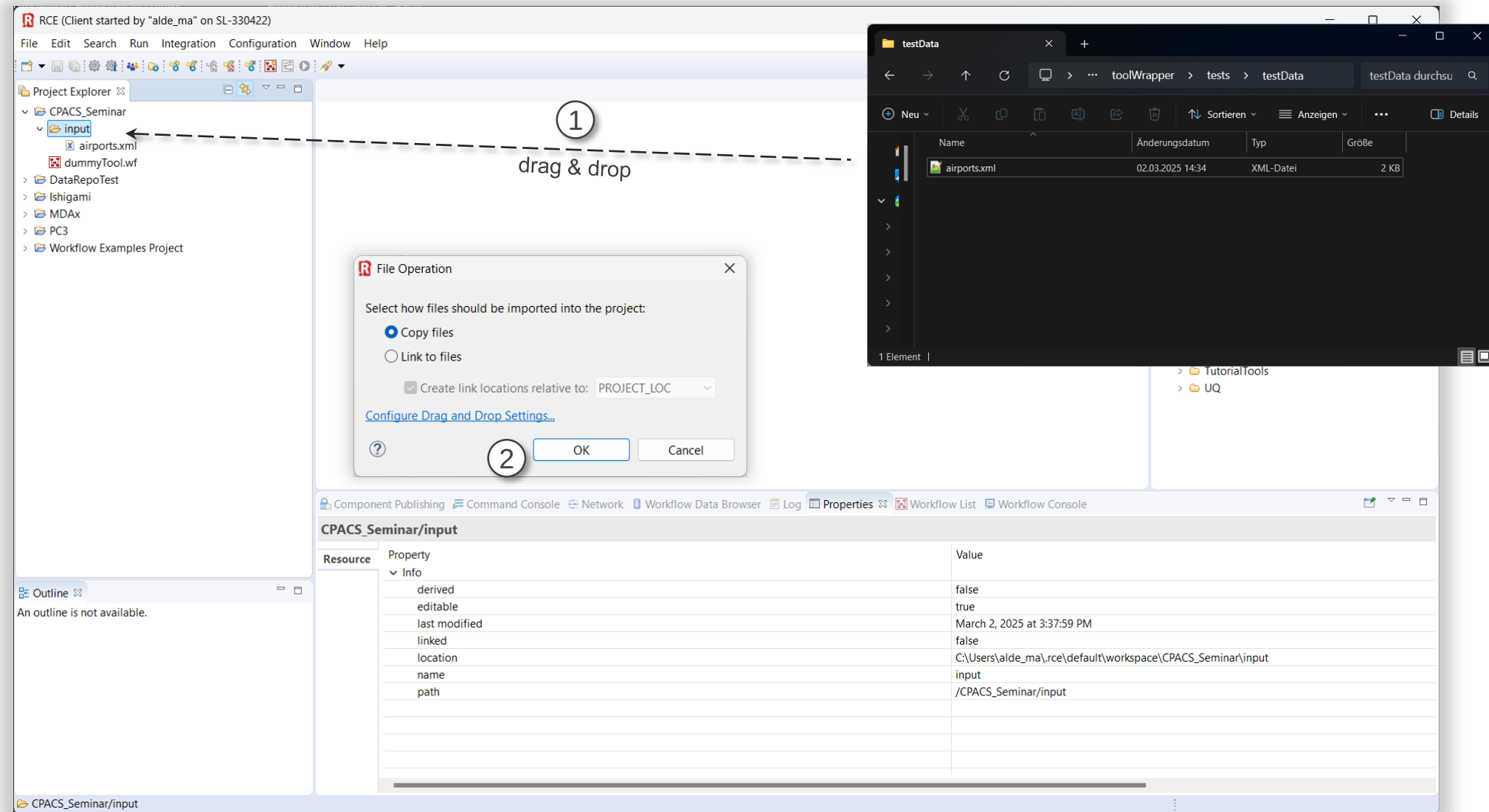
< Back Next > Save As ... Save and update Cancel

(2) "Save and update/activate"







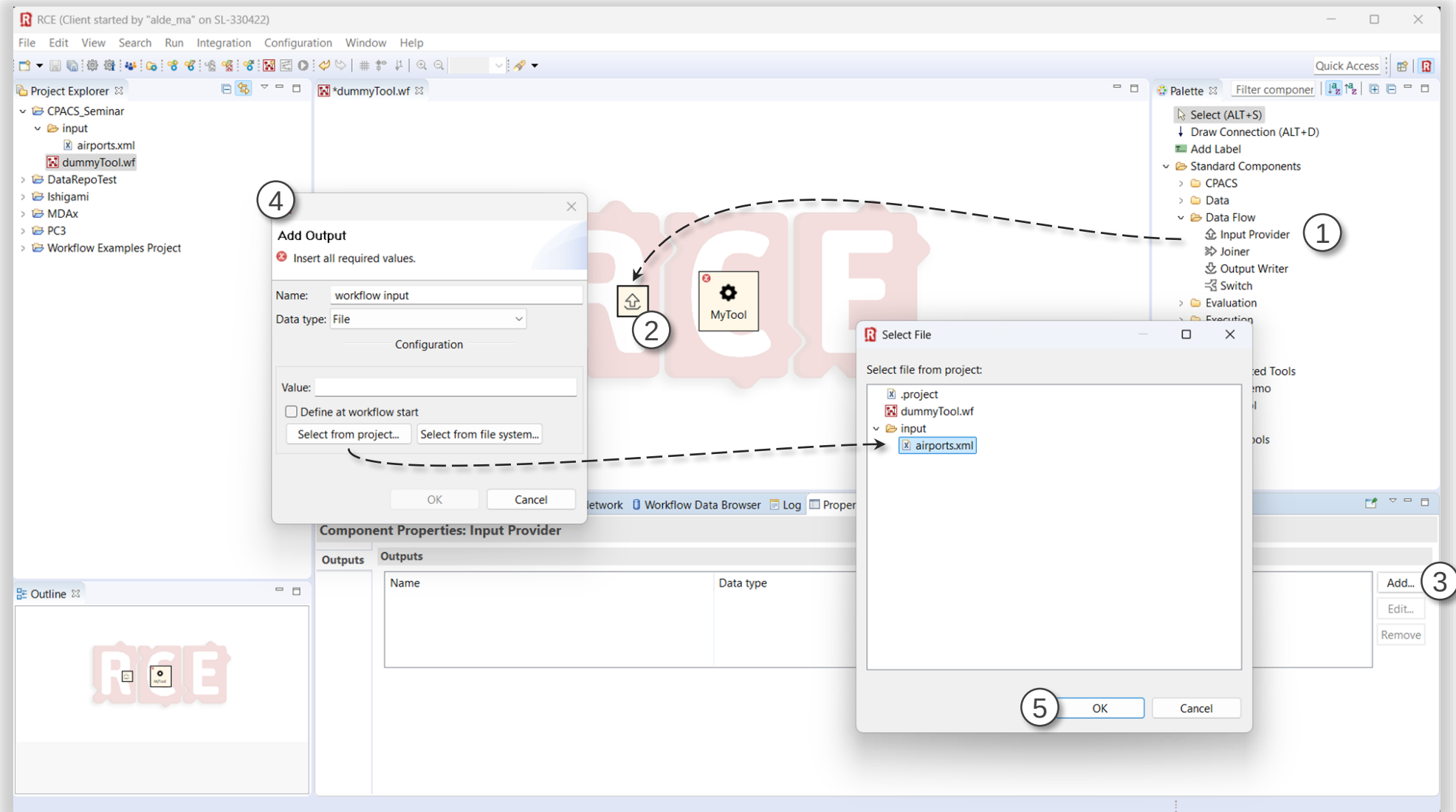


The screenshot shows the RCE (Client started by "alde_ma" on SL-330422) interface. The Project Explorer on the left shows a project structure with a folder named "input" under "CPACS_Seminar". A dashed arrow labeled "1" and "drag & drop" points from the "input" folder to a "File Operation" dialog box. The dialog box has two options: "Copy files" (selected) and "Link to files". Below these options is a checkbox "Create link locations relative to:" with a dropdown menu set to "PROJECT_LOC". A link "Configure Drag and Drop Settings..." is also present. The "OK" button is circled with a "2".

In the background, a file explorer window shows the contents of the "testData" folder, which includes a file named "airports.xml".

The Properties panel at the bottom right shows the properties for the "CPACS_Seminar/input" resource. The table below lists the properties and their values:

Property	Value
derived	false
editable	true
last modified	March 2, 2025 at 3:37:59 PM
linked	false
location	C:\Users\alde_ma\rce\default\workspace\CPACS_Seminar\input
name	input
path	/CPACS_Seminar/input



RCE (Client started by "alde_ma" on SL-330422)

File Edit View Search Run Integration Configuration Window Help

100%

Project Explorer

- CPACS_Seminar
 - input
 - airports.xml
 - dummyTool.wf
 - DataRepoTest
 - Ishigami
 - MDAx
 - PC3
 - Workflow Examples Project

dummyTool.wf

Palette

Filter componer

1

Select (ALT+S)

Draw Connection (ALT+D)

Add Label

Standard Components

- CPACS
- Data
- Data Flow
 - Input Provider
 - Joiner
 - Output Writer
 - Switch
- Evaluation
- Execution
- XML
- _Internal
- User Integrated Tools
 - CPACS_demo
 - MyTool
 - DLR-SL
 - TutorialTools
 - UQ

2

3

Connection Editor

Source

- Input Provider
 - workflow input

Connections

Target

- MyTool
 - XML_in

Note: One or more connections have ...n automatically. Click OK to apply.

Cancel OK

Component Publishing Command Console Network Workflow Data Browser Log Properties Workflow List Workflow Console

Component Properties: Input Provider

Outputs

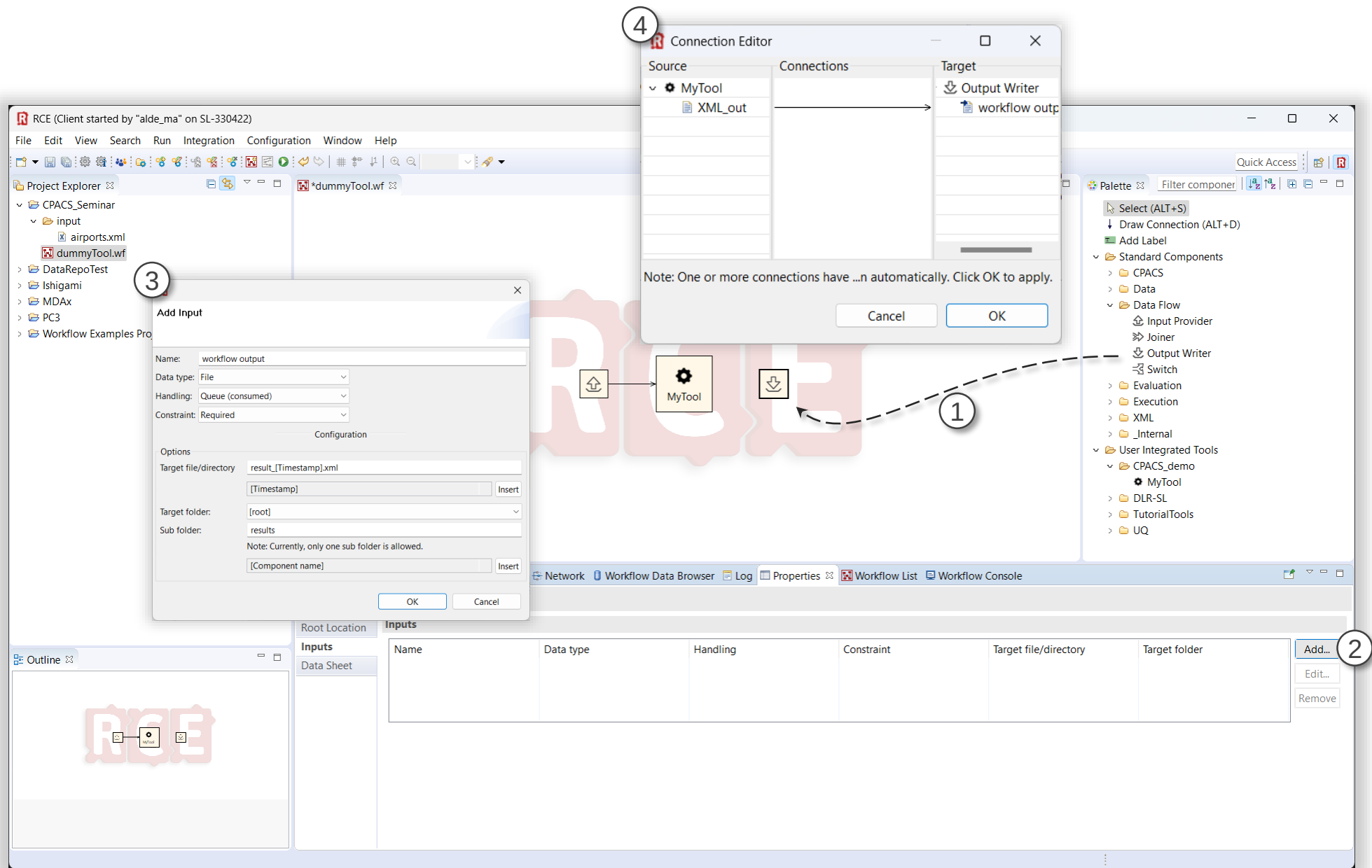
Name	Data type	Value
workflow input	File	CPACS_Seminar/input/airports.xml

Add... Edit... Remove

Note: You must run the workflow (containing this Input Provider component) from the RCE instance, which host stores the files and/or the directories defined.

Outline

RCE



The screenshot shows the RCE (Client started by "alde_ma" on SL-330422) interface. The main window displays a workflow diagram with a "MyTool" component. The "Add Input" dialog is open, showing the configuration for the "workflow output" input. The "Connection Editor" dialog is also open, showing the connection between "XML_out" and "workflow output". The "Palette" on the right lists various components, including "MyTool". The "Inputs" table at the bottom is empty, and the "Add..." button is highlighted with a circled 2.

Step 1: A dashed arrow points from the "Add Input" dialog to the "MyTool" component in the workflow diagram.

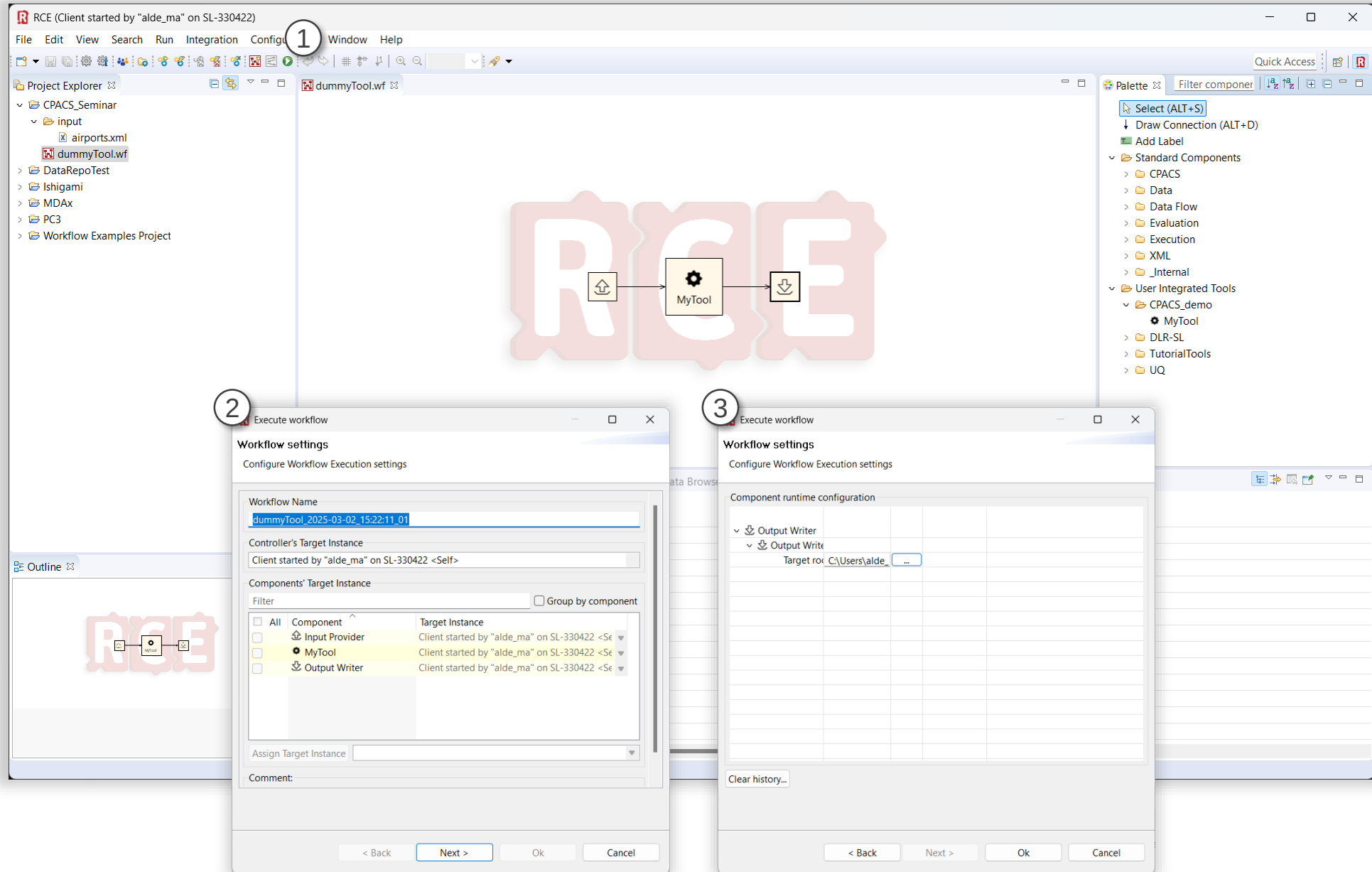
Step 2: The "Add Input" dialog is open, showing the configuration for the "workflow output" input. The "Name" field is "workflow output", "Data type" is "File", "Handling" is "Queue (consumed)", and "Constraint" is "Required". The "Options" section shows "Target file/directory" as "result_[Timestamp].xml", "Target folder" as "[root]", and "Sub folder" as "results".

Step 3: The "Add Input" dialog is open, showing the configuration for the "workflow output" input. The "Name" field is "workflow output", "Data type" is "File", "Handling" is "Queue (consumed)", and "Constraint" is "Required". The "Options" section shows "Target file/directory" as "result_[Timestamp].xml", "Target folder" as "[root]", and "Sub folder" as "results".

Step 4: The "Connection Editor" dialog is open, showing the connection between "XML_out" and "workflow output". The "Source" is "XML_out" and the "Target" is "workflow output".

Inputs Table:

Name	Data type	Handling	Constraint	Target file/directory	Target folder
------	-----------	----------	------------	-----------------------	---------------



The screenshot displays the RCE (Client started by "alde_ma" on SL-330422) interface. The main window shows a workflow diagram with three components: an input provider, a central tool named "MyTool", and an output writer. The workflow is titled "dummyTool.wf".

Two dialog boxes are overlaid on the main window, illustrating the configuration steps for executing the workflow:

- Execute workflow (Workflow settings):** This dialog box is used to configure the workflow execution settings. It includes fields for the Workflow Name (dummyTool_2025-03-02_15:22:11_01), the Controller's Target Instance (Client started by "alde_ma" on SL-330422 <Self>), and the Components' Target Instance (Client started by "alde_ma" on SL-330422 <Self>). A table lists the components and their target instances:

Component	Target Instance
Input Provider	Client started by "alde_ma" on SL-330422 <Self>
MyTool	Client started by "alde_ma" on SL-330422 <Self>
Output Writer	Client started by "alde_ma" on SL-330422 <Self>

- Execute workflow (Component runtime configuration):** This dialog box is used to configure the runtime settings for the components. It includes a table for the Component runtime configuration:

Component	Runtime Configuration
Output Writer	Output Write
Output Write	Target row: C:\Users\alde_...

RCE (Client started by "alde_ma" on SL-330422)

File Edit View Search Run Integration Configuration Window Help

Project Explorer

- CPACS_Seminar
 - input
 - airports.xml
 - results
 - result_2025-03-02_15-23-28-643.xml
 - dummyTool.wf
- DataRepoTest
- Ishigami
- MDAx
- PC3
- Workflow Examples Project

dummyTool.wf dummyTool_2025-03-02_15:23:08_03: Finished

Palette

- Select (ALT+S)
- Draw Connection (ALT+D)
- Add Label
- Standard Components
 - CPACS
 - Data
 - Data Flow
 - Evaluation
 - Execution
 - XML
 - _Internal
- User Integrated Tools
 - CPACS_demo
 - MyTool
 - DLR-SL
 - TutorialTools
 - UQ

Component Publishing Command Console Network Workflow Data Browser Log Properties Workflow List Workflow Console

CPACS_Seminar/results/result_2025-03-02_15-23-28-643.xml

Resource	Property	Value
Info	derived	false
	editable	true
	last modified	March 2, 2025 at 3:23:28 PM
	linked	false
	location	C:\Users\alde_ma\rce\default\workspace\CPACS_Seminar\results\result_2025-03-02_15-23-28-643
	name	result_2025-03-02_15-23-28-643.xml
	path	/CPACS_Seminar/results/result_2025-03-02_15-23-28-643.xml
	size	1,755 bytes

Outline

CPACS_Seminar/results/result_2025-03-02_15-23-28-643.xml

RCE (Client started by "alde_ma" on SL-330422)

File Edit View Search Run Integration Configuration Window Help

Project Explorer

- CPACS_Seminar
 - input
 - airports.xml
 - results
 - result_2025-03-02_15-23-28-643.xml ①
 - dummyTool.wf
- DataRepoTest
- Ishigami
- MDAx
- PC3
- Workflow Examples Project

dummyTool.wf dummyTool_2025-03-02_15:23:08:03: Finished

result_2025-03-02_15-23-28-643.xml

Node Content

header	all(name, description?, version, cpacsVersion?, ...)
name	Flight Path Example
version	1.0
versionInfos	(versionInfo+)
versionInfo	all(cpacsVersion, description, timestamp, creat...
version	1.0
cpacsVersion	3.5
description	Flight Path from Frankfurt to Newark
timestamp	2023-04-27T12:00:00
creator	Arthur Zamfir, DLR-SL
changeLog	(logEntry+)
logEntry	all(description, timestamp, creator)
logEntry	all(description, timestamp, creator)
description	Add my dummy tool data ②
timestamp	2025-03-02T15:23:27
creator	Dummy Tool
vehicles	all(aircraft?, rotorcraft?, engines?, performance...
airports	(airport+)

Design Source

Component Publishing Command Console Network Workflow Data Browser Log Properties Workflow List Workflow Console

CPACS_Seminar/results/result_2025-03-02_15-23-28-643.xml

Resource	Property	Value
Info	derived	false
	editable	true
	last modified	March 2, 2025 at 3:23:28 PM
	linked	false
	location	C:\Users\alde_ma\rce\default\workspace\CPACS_Seminar\results\result_2025-03-02_15-23-28-643
	name	result_2025-03-02_15-23-28-643.xml
	path	/CPACS_Seminar/results/result_2025-03-02_15-23-28-643.xml
	size	1,755 bytes

Outline

CPACS_Seminar/results/result_2025-03-02_15-23-28-643.xml

RCE (Client started by "alde_ma" on SL-330422)

File Edit View Search Run Integration Configur Window Help

Project Explorer

- CPACS_Seminar
 - input
 - airports.xml
 - results
 - result_2025-03-02_15-23-28-643.xml
 - dummyTool.wf
- DataRepoTest
- Ishigami
- MDAx
- PC3
- Workflow Examples Project

dummyTool.wf

Palette

Select (ALT+S)

Draw Connection (ALT+D)

Add Label

Standard Components

- CPACS
- Data
- Data Flow
- Evaluation
- Execution
- XML
- _Internal

User Integrated Tools

- CPACS_demo
 - MyTool
- DLR-SL
- TutorialTools
- UQ

Component Publishing Command Console Network Workflow Data Browser Log Properties Workflow List Workflow Console

Component Properties: MyTool (1)

Configuration

Working directory

Inputs/Outputs

Imitation Mode

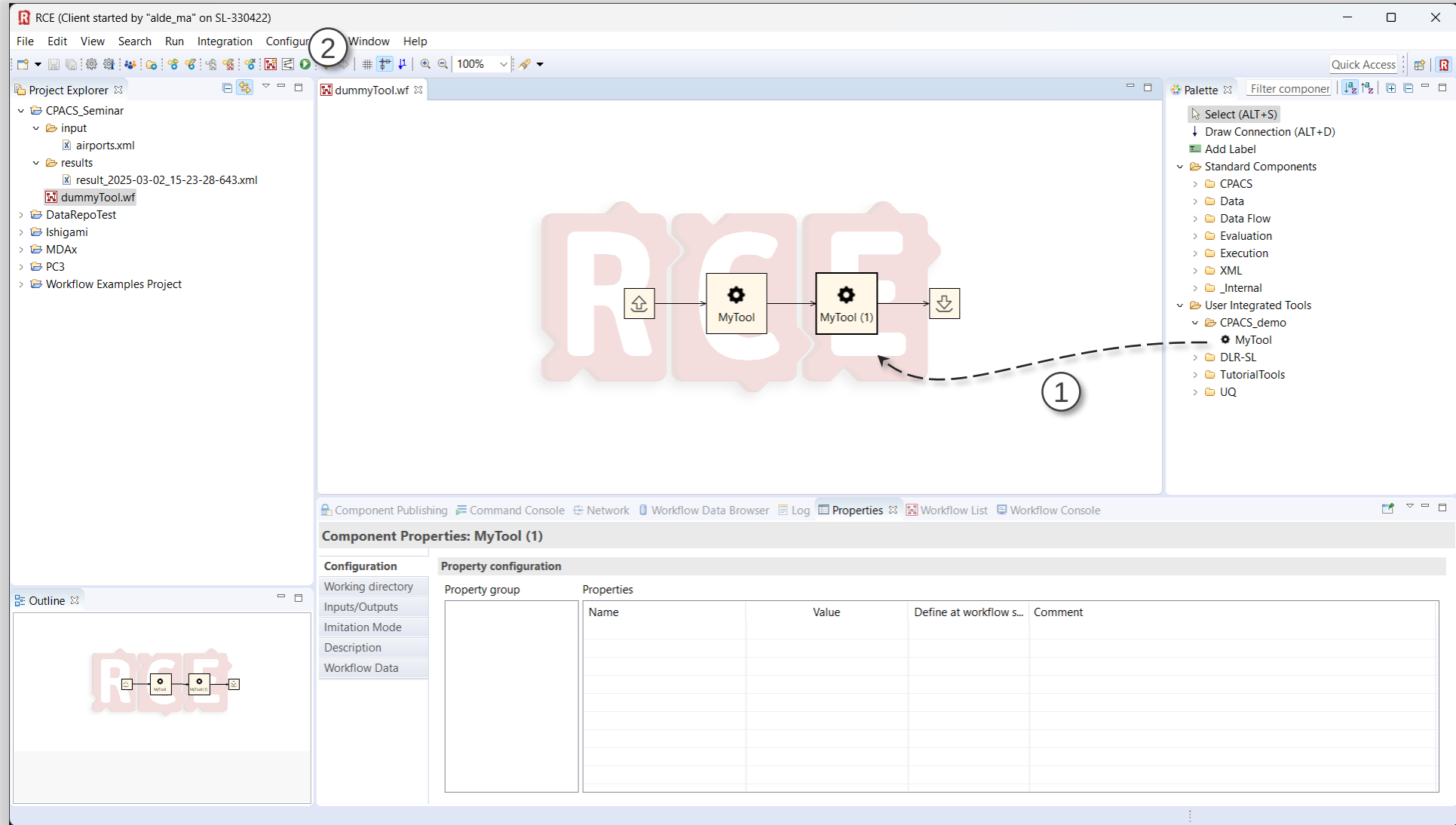
Description

Workflow Data

Property configuration

Property group	Properties			
	Name	Value	Define at workflow s...	Comment

Outline



RCE (Client started by "alde_ma" on SL-330422)

File Edit Search Run Integration Configuration Design Window Help

Project Explorer

- CPACS_Seminar
 - input
 - airports.xml
 - results
 - result_2025-03-02_15-23-28-643.xml
 - result_2025-03-02_15-26-52-175.xml
 - dummyTool.wf
 - DataRepoTest
 - Ishigami
 - MDAx
 - PC3
 - Workflow Examples Project

1

dummyTool.wf

dummyTool_2025-03-02_15:26:49_05: Finished

result_2025-03-02_15-26-52-175.xml

Node

Node	Content
versionInfos	(versionInfo+)
versionInfo	all(cpacsVersion, description, timestamp, creator)
version	1.0
cpacsVersion	3.5
description	Flight Path from Frankfurt to Newark
timestamp	2023-04-27T12:00:00
creator	Arthur Zamfir, DLR-SL
changeLog	(logEntry+)
logEntry	all(description, timestamp, creator)
logEntry	all(description, timestamp, creator)
logEntry	all(description, timestamp, creator)
logEntry	all(description, timestamp, creator)
vehicles	all(aircraft?, rotorcraft?, engines?, performance?)

2

Component Publishing Command Console Network Workflow Data Browser Log Properties Workflow List Workflow Console

Property

Property	Value
externalDataDirectory	
externalDataNodePath	
externalFileName	

Outline

- versionInfos
 - versionInfo version=1.0
 - cpacsVersion
 - description
 - timestamp
 - creator
 - changeLog
 - logEntry
 - logEntry
 - logEntry

cpacs/header/versionInfos/versionInfo/changeLog/logEntry

Writable Smart Insert 1:1

Topic: **RCE Tool Integration**
Dummy Tool wrapper

Date: 2025-03-02

Author: CPACS Team

Institute: DLR Institute for System Architectures in Aeronautics

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